

Layout commands

These helpers make poster geometry explicit.

- `pframe`: one Beamer frame plus left/right gutters.
- `pcolumn{width}`: a top-aligned minipage of exact width.
- `prow`: puts columns on one row; use it for nested columns.
- `pblock[style]{title}{height}`: block with a target height.

Use `alert` or `example` as optional block styles. If content is taller than the target, the block grows instead of overprinting the next block.

Usual skeleton

`pframe` → `pcolumn` → `pblock`

Nested layout: `prow` → `pcolumn` → `pblock`.

Height recipe

Give every block a target height and let the last block fill the remainder.

$$\sum h_i + (M - 1) \backslash \text{blocksepheight} = \backslash \text{textheight}$$

For a column with blocks of 24cm, 12.5cm, and one remaining block:

$$\backslash \text{textheight} - 24\text{cm} - 12.5\text{cm} - 2\backslash \text{blocksepheight}$$

Use	Pattern
Same-height stack	choose fixed cm values
Fill leftover space	subtract used heights from <code>\textheight</code>
Two-column span	<code>2\colwidth + \pcolumnsepwidth</code>
Too much content	block grows; shorten text or increase height

Keep these formulas near the length definitions in the preamble.

Width recipe

Pick the number of columns first, then solve for widths.

- Top level: $(N + 1)s + Nw = \backslash \text{textwidth}$.
- Nested row: $\sum w_i + (N - 1)s = \text{parent width}$.
- Here: $s = \backslash \text{pcolumnsepwidth}$ and $w = \backslash \text{colwidth}$.

Changing type size

Global sizes live in `beamerthemerpi.sty`.

- `block title`: usually `\Large` and bold.
- `block body`: usually `\large`.
- `headline title`: usually `\Huge`.

Local override: `{\Large ...}`. Exact size: `\fontsize{34}{40}\selectfont`.

Changing separator lengths

Two lengths control most whitespace.

- `\pcolumnsepwidth`: horizontal gutter between columns.
- `\blocksepheight`: vertical gap between stacked blocks.

Set them in the preamble:

`\setlength{\pcolumnsepwidth}{0.025\textwidth}`

`\setlength{\blocksepheight}{4ex}`

Mini layout checklist

1. Choose poster size in `beamerposter`.
2. Choose `\pcolumnsepwidth`.
3. Solve for `\colwidth`.
4. Set span widths, e.g. `\twocolwidth`.
5. Pick block heights to sum to `\textheight`.
6. Build, then trim text or adjust heights.

Rule of thumb

Make the grid first, then write to fit it. Clean posters have shared gutters, aligned bottoms, and short text.

Why not Beamer columns?

Beamer's columns is made for slides, not grid-filling posters.

- It adds internal spacing, so the visible width is not just the number you typed.
- Column height follows content; it does not reserve an exact block height.
- Nested layouts are harder to make flush with the outer grid.

The custom helpers use plain minipages and fixed gutters, so the math stays predictable.

References

- [1] Claude E. Shannon.
A mathematical theory of communication.
Bell System Technical Journal, 27(3):379–423, 1948.